



TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN, ĐHQG-HCM
ĐỀ THI KẾT THÚC HỌC PHẦN
Học kỳ II - Năm học 2024 - 2025

MÃ LƯU TRỮ
(do phòng KT-DBCL ghi)
CK2425-2
STAT451

Tên học phần: Applied Statistics for Engineering and Scientists 1 Mã HP: STAT451
Thời gian làm bài: 90 phút (23TT1 + 23TT2) Ngày thi: 21/4/2025
Ghi chú: Sinh viên [☐ được phép / ☒ không được phép] sử dụng tài liệu khi làm bài.

Họ tên sinh viên: Nguyễn Hoàng Gia Hân.....MSSV: 23145055.....STT:.....

Note:

- All calculations are rounded to four decimal places.
- The standard normal distribution $N(0,1)$ and Student's t distribution tables can be found in Appendices 1 and 2 on pages 3 and 4, respectively.

Question 1. (2.0 points) A forester studying the effects of fertilization on certain pine forests in the southeast of a country is interested in estimating the average basal area of pine trees. Based on years of studying the basal areas of similar trees, he has found that these measurements (in square inches) are normally distributed, with a standard deviation of approximately 4 square inches. ♂

- (a) If the forester samples $n = 9$ trees, what is the probability that the sample mean will be within 2 square inches of the population mean?
- (b) Suppose the forester wants the sample mean to be within 1 square inch of the population mean with a probability of 0.90. How many trees must he measure to ensure this level of accuracy?

Question 2. (2.0 points) For developing countries in Africa and the Americas, let p_1 and p_2 be the respective proportions of babies with a low birth weight (below 2500 grams). We will test the null hypothesis $H_0 : p_1 = p_2$ against the alternative hypothesis $H_1 : p_1 > p_2$. Random samples of sizes $n_1 = 900$ and $n_2 = 700$ yielded $y_1 = 135$ and $y_2 = 77$ babies with a low birth weight.

- (a) What is your decision at a significance level of $\alpha = 0.01$?
- (b) What is the p -value of the test?

Question 3. (2.0 points) The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

Drying time (hours)	2.8	3.0	3.2	3.4	3.6	3.8
Number of paints	20	36	48	33	18	14

Assume that the measurements represent a random sample from a normal distribution.

- (a) Construct a 95% confidence interval for the mean drying time.
- (b) There is a claim that the average drying time for paints that take more than 3.3 hours to dry is 3.58 hours. Test this claim at the 5% significance level. ✓
- (c) Construct a 99% confidence interval for the proportion of paints with a drying time of at least 3.3 hours.

Họ tên người ra đề/MSCB: Chữ ký: (Đề thi gồm 4 trang)
[Trang 1/4]
Họ tên người duyệt đề: Chữ ký:

Question 4. (1.5 points) A study was conducted to compare the average heat retention time of two different brands of thermal flasks. Fourteen units of Brand X were tested, each filled with hot water and monitored until the temperature dropped below a standard threshold. Similarly, twelve units of Brand Y were tested. The heat retention time (in hours) was recorded for each unit.

The sample for Brand X had an average heat retention time of 27.3 hours, with a sample standard deviation of 1.8 hours. The sample for Brand Y had an average heat retention time of 24.6 hours, with a sample standard deviation of 2.1 hours.

Can we conclude, at the 0.05 level of significance, that the average heat retention time of Brand X exceeds that of Brand Y by more than 2 hours? Assume that the population distributions are approximately normal with equal variances.

Question 5. (2.5 points) The ages (in years) of 10 infants and the number of hours each slept in a day are shown in the table below:

Age (years)	0.1	0.2	0.4	0.7	0.6	0.9	0.1	0.2	0.4	0.9
Hours slept	14.9	14.5	13.9	14.1	13.9	13.7	14.3	13.9	14.0	14.1

- Determine the linear correlation coefficient between Age and Hours slept. Interpret the result.
- A researcher wants to predict the number of hours an infant sleeps per day based on age. Find the equation of the linear regression line for the given data. Interpret the estimated coefficients.
- Use the linear regression equation obtained in part (b) to predict the number of hours slept for each of the following ages, if appropriate. If an age is not appropriate for prediction, explain why.
 - Age = 0.3 year
 - Age = 3.9 years
 - Age = 0.6 year
 - Age = 0.8 year
- Determine the R^2 value of the linear model in part (b). Interpret the result.